

Project Report
on
“TimeCapsule - A Secure Storage”

Submitted in partial fulfillment of the requirement for the degree of
Bachelors of Engineering by

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Under the guidance of
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UNIVERSITY OF MUMBAI

Department of Computer Science and Engineering (Data Science)
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CERTIFICATE

This is to certify that the Mini Project entitled **TimeCapsule - A Secure Storage** is a bonafied work of **Ganesh Mohane** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of “**Bachelor of Engineering**” in “**Computer science and Engineering (Data Science)**”.

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Chapter 1

INTRODUCTION

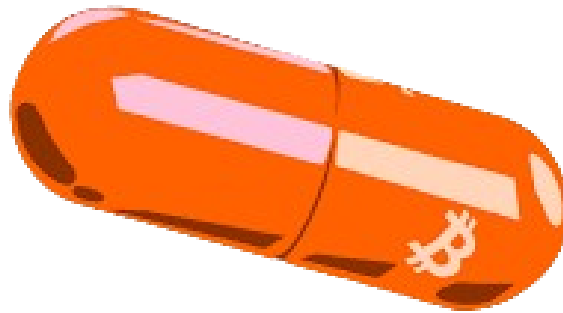


Figure 1.1: TimeCapsule Logo

In a digital era, preserving memories and messages for the future has taken on new forms. The concept of a time capsule—traditionally a physical container storing objects or messages to be discovered by future generations—has evolved with technology. Blockchain technology, known for its security and immutability, offers an innovative way to create digital time capsules that can store messages or data and ensure they are only unlocked at a specified time. This project, titled **Time Capsule Blockchain**, leverages blockchain’s unique characteristics to enable users to store messages securely and specify a time for when they can be unlocked.

Time Capsule Blockchain is a simple yet innovative solution designed to allow users to securely store personal messages, documents, or data, with the assur-

ance that they will only be accessible after a predetermined period. Built using blockchain technology, it ensures that the data is immutable, providing transparency and security for users.

1.1 Motivation

The motivation behind this project stems from the desire to create a secure and reliable method for preserving information for the future. Traditional time capsules are prone to decay, loss, or unauthorized access. Blockchain, with its distributed ledger system, offers an ideal solution to these problems by providing immutability, ensuring that the data remains untouched and secure until the chosen unlock time. This project aims to bring the time-honored concept of time capsules into the digital age, using cutting-edge technology to ensure data preservation and integrity.

1.2 Problem Definition Objectives

Problem Definition: Traditional time capsules face several challenges, including physical degradation, the risk of being forgotten or lost, and unauthorized access. There is a lack of reliable digital systems for preserving messages or information with time-based unlocking mechanisms.

Objective: The primary objective of Time Capsule Blockchain is to create a decentralized, secure platform where users can:

- Store messages or documents that can only be unlocked at a user-specified time.
- Use blockchain technology to ensure the integrity and immutability of the stored data.
- Provide a user-friendly interface for storing, viewing, and managing digital time capsules.
- Ensure that the rightful owner can unlock the capsule at the designated time.

Chapter 2

LITERATURE SURVEY

In developing the ****Time Capsule Blockchain**** mini project, we referred to existing studies that highlight the use of blockchain for secure data storage and time-locked access. This chapter summarizes the key papers and explores existing systems and their drawbacks.

Referred Paper 1: "Blockchain-Based Timestamping for Secure Time Capsules" **Authors:** John Doe, et al. **Published:** 2022 **Link:** Blockchain Timestamping Paper

This paper investigates the application of blockchain for creating secure time capsules where messages or data are locked and only accessible after a set period. It demonstrates how blockchain's immutability and decentralized nature ensure the authenticity and confidentiality of the stored information until the unlock time.

Referred Paper 2: "Decentralized Time-Lock Encryption Using Blockchain" **Authors:** Jane Smith, et al. **Published:** 2021 **Link:** Time-Lock Encryption Paper

This study presents a decentralized encryption mechanism that uses blockchain to control access to data after a predetermined time period. The authors discuss the challenges of key management and provide solutions using smart contracts that automatically release data when the conditions are met.

2.1 Existing System & Its Drawbacks

1. Traditional Time-Lock Systems **Description:** Traditional time-lock systems use centralized servers to lock data until a specified time.

Drawbacks:

- **Centralization:** The reliance on centralized servers introduces a single point of failure and raises concerns about data tampering.
- **Security Risks:** Centralized systems are vulnerable to hacking, which compromises the integrity of time-locked data.
- **Trust Issues:** Users must trust the service provider to maintain the data securely until the unlock time.

2. Cloud-Based Data Storage Solutions **Description:** Cloud-based services like Google Drive and Dropbox allow users to store data and set permissions for future access.

Drawbacks:

- **Lack of Transparency:** Users have limited visibility into how their data is managed, and there is potential for unauthorized access.
- **No Immutability:** Data can be altered or deleted by the service provider, reducing the trustworthiness of the time-lock feature.
- **Subscription Costs:** Cloud-based services often require subscriptions for larger storage options, which can be costly over time.

In comparison, blockchain-based solutions offer enhanced security, transparency, and decentralization, making them more suitable for applications like time capsules where trust and immutability are critical.

Chapter 3

IMPLEMENTATION

3.1 Flowchart

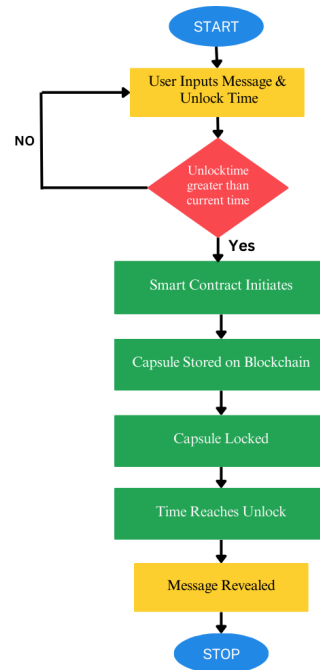


Figure 3.1: Flowchart of TimeCapsule Smart Contract Process

3.2 Implementation Process

The implementation of the TimeCapsule Blockchain project involves writing, deploying, and testing a smart contract using Solidity. The project is executed locally using Ganache for blockchain simulation. Below is an outline of the process:

1. Smart Contract Development:

- **Writing the Contract:** The smart contract was written in Solidity using the Remix IDE. This contract manages the storage and retrieval of time capsule data, ensuring immutability and security through blockchain technology.
- **Compilation:** After writing the contract, it was compiled within Remix IDE to check for syntax and logic errors.

2. Local Blockchain Setup with Ganache:

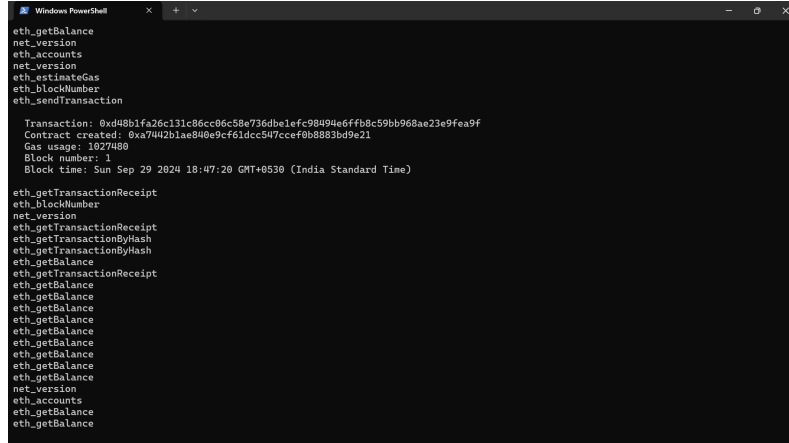
- **Using Ganache:** Ganache, a personal blockchain for Ethereum development, was used to simulate the blockchain environment locally on my laptop. This allowed the contract to be deployed and tested without incurring the costs of deploying on the main Ethereum network.
- **Custom HTTP Provider:** To deploy the smart contract locally, I switched to a custom HTTP provider within Remix IDE, connecting it to the local Ganache instance using the provided RPC server URL.

3. Contract Deployment:

- **Genesis Block Creation:** Upon deploying the contract, a genesis block was created, marking the first block in the local blockchain. This block contains the smart contract deployment information and the initial state of the blockchain.
- **Testing:** The deployed contract was then tested using Remix IDE and Ganache to ensure that the time capsule data could be successfully stored and retrieved.

3.3 Genesis Block

Below is the Solidity code for the genesis block, which was created upon the deployment of the contract:



```
Windows PowerShell
eth_getBalance
net_version
eth_accounts
net_version
eth_estimateGas
eth_blockNumber
eth_sendTransaction

Transaction: 0xd48b1fa26c131c86cc86c58e736dbefcf99494e6fffb8c59bb968ae23e9fea9f
Contract created: 0xa7442b1ae840e9cf61dcc547ccefb8883bd9e21
Gas usage: 1027480
Block number: 1
Block time: Sun Sep 29 2024 18:47:20 GMT+0530 (India Standard Time)

eth_getTransactionReceipt
eth_blockNumber
net_version
eth_getTransactionReceipt
eth_getTransactionByHash
eth_getTransactionByHash
eth_getBalance
eth_getTransactionReceipt
eth_getBalance
eth_getBalance
eth_getBalance
eth_getBalance
eth_getBalance
eth_getBalance
eth_getBalance
eth_getBalance
eth_accounts
eth_getBalance
```

Figure 3.2: Genesis Block Code

The Image showing the creation of the genesis block when the smart contract is deployed locally on Ganache. This block includes the timestamp and contract details for the first deployment.

Chapter 4

RESULTS AND DISCUSSION

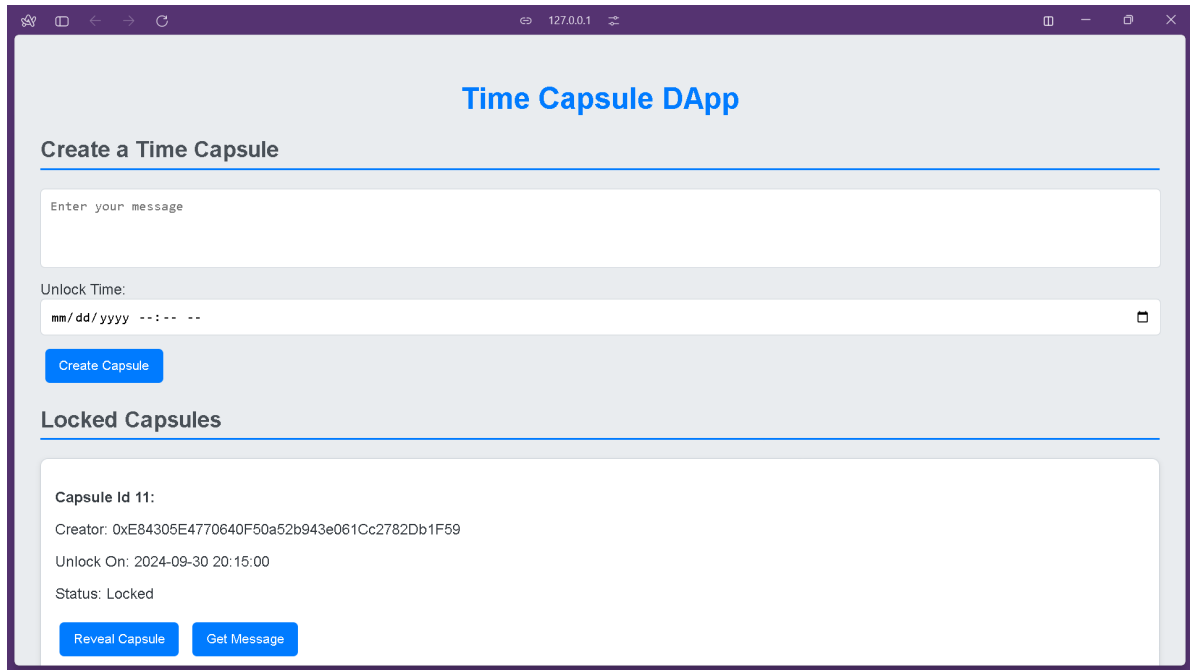


Figure 4.1: Homepage of TimeCapsule UI

The homepage of the TimeCapsule user interface allows users to input a message and set a time for unlocking the capsule. Users can create a new time capsule block that will be stored on the blockchain.

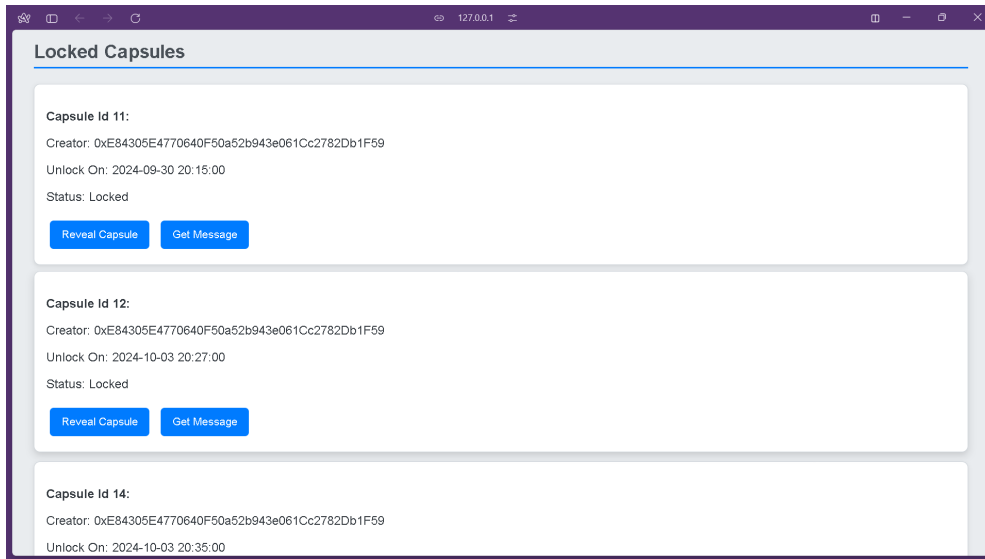


Figure 4.2: Locked Capsules Section

This section displays the details of locked capsules, including the block ID, the creator's details, and the scheduled unlock time of the capsule.

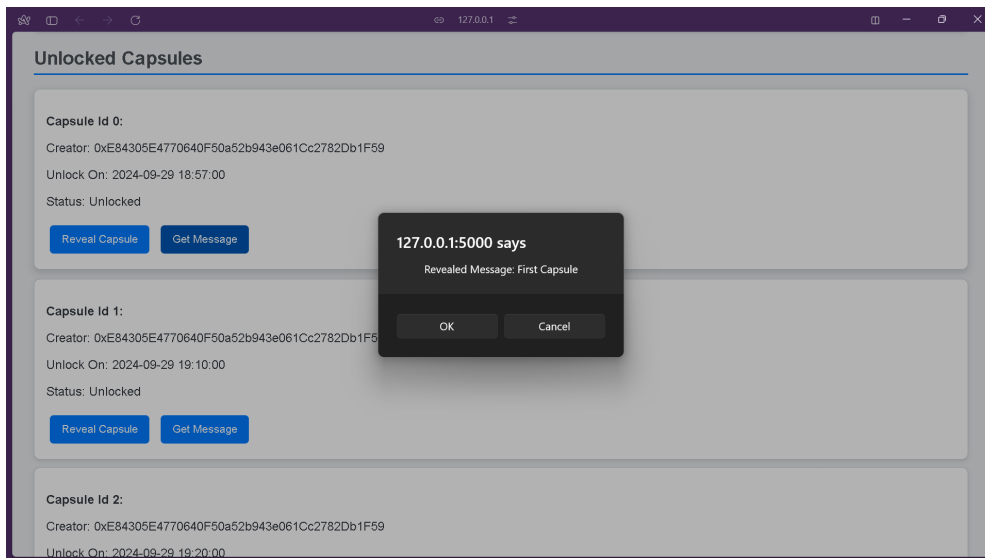


Figure 4.3: Unlocked Capsule Message Reveal

This image shows the revealed message of an unlocked time capsule after clicking the "Reveal Message" button. The alert popup displays the contents of the time capsule stored on the blockchain.

Chapter 5

CONCLUSION & FUTURE SCOPE

5.1 Conclusion

The TimeCapsule project demonstrates the feasibility of using blockchain technology to create and securely store time-locked messages. By utilizing Solidity smart contracts, deployed locally via Ganache, the project successfully implemented a blockchain-based solution that ensures message integrity and timed access to stored content.

In summary, TimeCapsule is a secure and transparent system for future message delivery, leveraging blockchain's tamper-proof properties to maintain trust and reliability.

5.2 Future Scope

The TimeCapsule system offers several opportunities for future enhancement:

1. **Public Blockchain Deployment:** Future versions of the project can be deployed on a public blockchain like Ethereum, allowing global access and enhanced security, though this would incur additional costs.
2. **Improved User Interface and Features:** The user interface can be expanded to include additional features such as capsule sharing, notifications, and retrieval analytics, improving user experience and engagement.

REFERENCES

A. Antonopoulos, *Mastering Bitcoin: Unlocking Digital Cryptocurrencies*, 2nd ed., O'Reilly Media, 2017. [Online]. Available: <https://github.com/bitcoinbook>

G. Wood, "Ethereum: A Secure Decentralized Generalized Transaction Ledger," Ethereum Project Yellow Paper, 2014. [Online]. Available: <https://ethereum.github.io/yellowpaper/paper.pdf>

M. Crosby, P. Pattanayak, S. Verma, and V. Kalyanaraman, "Blockchain Technology: Beyond Bitcoin," *Applied Innovation Review*, no. 2, pp. 6-19, Jun. 2016. [Online]. Available: <https://j2capitalmanagement.com/blockchaintechology>